

Chapter 1

Methodology

In the following chapter, I would like to highlight my testing procedure, but more importantly the reasoning behind why I make the specific choices that I do as it pertains to experimental setup. The aim of the experiments that are detailed in this chapter is to validate theoretical work, and to ensure that the theoretical development is practically achieved in the real world.

1.1 Experimental Design

1.1.1 Overview

The overall workings of the RADAR system can be split into three distinct sections of operation. The first section involves the ability of the system to act as a communication system. The second section of operation is the ability of the RADAR to detect objects at different ranges. The third area of operation of the RADAR is the ability of the RADAR to perform target classification. Due to this separation of operations, the testing of the system has been divided into three separate testing sections. The various testing categories are listed below.

1. **Communication Testing:** This test is all about how well the system can operate as a bare-bones communication system. The reason I have decided to include this as a test is because, as mentioned before, my entire classification pipeline is built upon OFDM communication principles (namely the use of pilot symbols in determining channel estimation data), and since these OFDM principles are so vital to my project, it is fitting to make sure that the system can actually operate as a data transmission system. This is necessary to show, since the broad category of RADAR that this system falls into could be considered ‘communication RADAR’, or in essence, a RADAR that can use the same signal within communication operations and normal RADAR operations.
2. **Range Testing:** The purpose of the range test will be to determine if the RADAR system can operate as just that, a simple RADAR. The main aim is to determine if the system can detect objects at various distances from the transmitter and receiver setup.
3. **Classification Testing:** This is the main testing that will take place for my RADAR system. This testing setup will thoroughly analyze the ability of the RADAR system to classify various targets in a scene. The tests will also try to determine the viability of this concept in a real world use case scenario.

Following on from the above listed broad testing structures, the exact testing procedure of each testing category will now be explained below.

1.1.2 Communication Test

The communication testing procedure will be paramount in determining if my system is working properly as a communication system, and therefore it is important to check the communication ability to determine if the calculated channel responses are valid.

As mentioned in chapter 3, one frame of my OFDM based signal is 32768 samples long. That means that there are 4384 samples either side of my ‘signal block’ that have values equal to zero. These are my guard bands. That leaves 24000 samples for data and pilot symbols. I made my signal have 4000 pilots and 20000 data carriers. Based on this setup, I can transmit 40000 bits at a time. On the reception side, I am able to read ingest raw data, which I can then decode and get my transmitted bits out of. Bearing this in mind, since I have the bits that I transmitted, and the bits that I have received, I am able to compute a BER for the transmission. The performance of the system changes drastically as the overall [Signal to Noise Ratio \(SNR\)](#) of the transmission changes.

To try and manipulate the SNR of the transmission, I want to perform multiple tests at various amplification levels. At severely low amplification levels, one can deduce that there will be very little signal being transmitted through the system.

1.1.2.1 Test 1: Communication System Characterization via Loopback

Since my communication tool chain has the ability to calculate the SNR of each and every capture that it processes, test 1 will work with this understanding. I wanted to design a test that shows, as good as possible, the overall communication system performance. One way that this can be shown, is by measuring the performance of the system under different stress conditions. So, for a communication system, that would involve measuring the performance of the system under varying SNR conditions. A varying SNR is a good choice of system stress conditions to determine real world raw communication performance. This is because it simulates the real world conditions that most signals (at least those operating in a free spectral band) have to deal with.

Unfortunately, there is no way to directly control the SNR in a real world test like this, as opposed to a simulation. Therefore, it is necessary to control another aspect of the signal which directly affects the SNR. In this test’s case, the best way to do that is to control the level above the noise floor that the signal exists on the received side of the system. The best way to achieve this, is to attach an attenuator between the transmit and receive ports of the system. Then, from there, one can adjust the level of attenuation, which in turn, raises and lowers the level that the transmitted signal sits above the noise floor.

In my experience with the USRP-N210 paired with the SBX daughterboard, the noise floor tends to hover around -100 dBV. The minimum level of attenuation that I have been using to test my system is roughly 15dB. Therefore, for this test of the communication, I will transmit the same communication signal through an attenuator, and step the attenuator from 100dB to 15dB of attenuation.

I have access to the following attenuators in my capacity as a UCT student, the Keysight 8496B, and the Keysight 8494B. These are pictured in figure 1.1 below:



(a) Keysight 8496B



(b) Keysight 8494B

Figure 1.1: Keysight Attenuators

While the attenuators pictured above come in the same package, they are anything but the same. The differences are listed below.

	Keysight 8496B	Keysight 8494B
Frequency Range	DC to 18 GHz	DC to 18 GHz
Steps	10 dB	1 dB
Attenuation Range	0 dB - 110 dB	0 dB - 11 dB

Table 1.1: Keysight 8496B and 8494B Attenuator Characteristics

Importantly, both attenuators operate from 0 Hz to 18 GHz, which is more than enough bandwidth for testing this system, since this system is limited to operations within the bandwidth of the SBX daughterboard of the USRP-N210, which can transmit signals on a carrier wave of 400 MHz to 4.4 GHz. More over, during the system design, I settled on doing tests in the portion of the spectrum from around 2 GHz to 3 GHz.

Using both attenuators, daisy-chained together, gave me the ability to control attenuation in the range from 0 dB to 121 dB in steps of 1 dB. The setup for this loopback test can be seen in the diagram presented in fig 1.2 below:

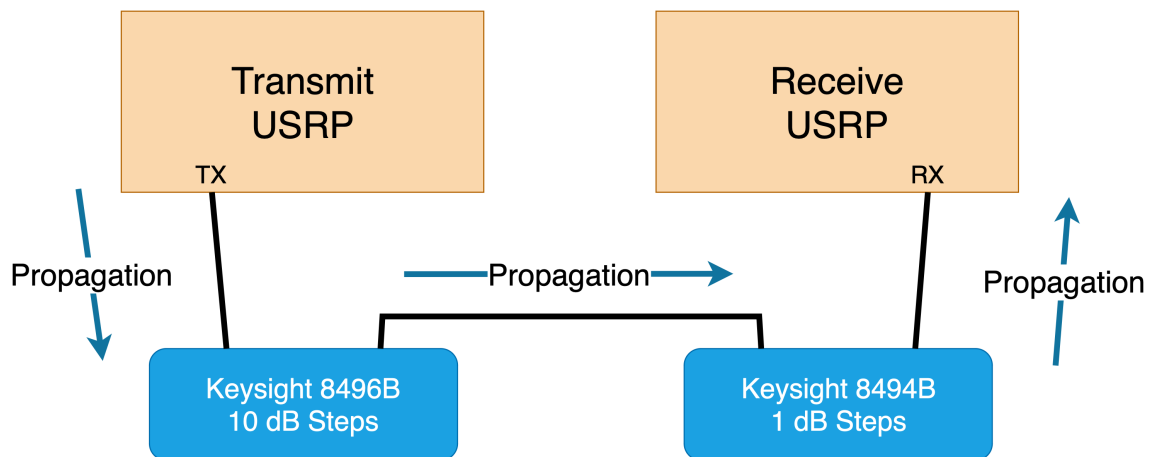


Figure 1.2: Communication Loopback Test

With this testing setup, as mentioned before, I have control over the attenuation of the overall signal propagation, effectively giving me indirect control over how high above the noise floor the transmitted signal sits, or in other word, the SNR. For the test, I will transmit a known communication signal, with known bits embedded into the signal. On the reception side, I can get a channel estimation, equalize the signal and demodulate the signal to extract data bits from the signal. I will do this for each attenuation level, varying from 100 dB to 15 dB of attenuation. For each signal that is received and demodulated, I am able to get an SNR and a BER. To ensure thorough characterization of the communication tool chain, I will perform 50 transmissions for each attenuation level. I will take all 50 SNR and BER values per attenuation level, and average them. This should give some very good insight into the performance of the system. The raw data format of the extracted data from this test will be in the format as shown below in table 1.2:

Attenuation	SNR_1 (dB)	BER_1 (%)	.	SNR_{50} (dB)	BER_{50} (%)	$Avg.SNR$ (dB)	$Avg.BER$ (%)
100 dB	x_1	y_1	.	x_{50}	y_{50}	.	.
99 dB	x_1	y_1	.	x_{50}	y_{50}	.	.
.	x_1	y_1	.	x_{50}	y_{50}	$\bar{x} = \frac{1}{n} \sum_{i=1}^{50} x_i$	$\bar{y} = \frac{1}{n} \sum_{i=1}^{50} y_i$
16 dB	x_1	y_1	.	x_{50}	y_{50}	.	.
15 dB	x_1	y_1	.	x_{50}	y_{50}	.	.

Table 1.2: Communication Test 1 Data Format

Something that must be noted however, is that since my system calculates an SNR value for a received signal after said signal has propagated through a noisy environment, it will struggle to measure the true SNR of the signal below a certain attenuation level. My system will not be able to truly calculate the SNR from the received signal when the embedded signal is below the noise floor. This is not such a large problem however, since one can determine the SNR values achieved for low values of SNR by extrapolating the values at high SNR values to the low SNR values. This can only be done because we are in complete control of the attenuation level of the loopback system.

1.1.2.2 Test 2: Long Cable Loopback Test

After completing a complete communication tool-chain characterization test as outlined in the previous section, it is now important to show that the system can actually perform it's required tasks in the real world. This entails two more tests. However, it must be noted that the next two tests will not be as thorough as test 1, since test 1 mainly dealt with characterization of the communication system. Test 2 and 3 will primarily to ascertain whether or not the system can operate under non-ideal (real-world) conditions. Test 2 will be another loopback test, however this one will make use of a particularly long coaxial cable. This allows one to examine the performance of the communication system in a slightly more real world operational conditions, namely, the fact that the coaxial is exposed to more real-world environmental noise since it is longer than the shorter loopback test.

1.1.2.3 Test 3: Real World Antenna Test

The third and final test will involve a demonstration that the system can work while transmitting data over a real-world channel, namely, that of the air between two separate antennas. The goal with this

test, much like the second test, will be to demonstrate that the system can indeed work in non-ideal conditions.

1.1.3 Range Test Tests

1.1.4 Classification Test

[1]

Bibliography

- [1] Christian Wolff. Classification of radar systems (1). [Online]. Available: <https://www.radartutorial.eu/02.basics/Classification%20of%20Radar%20systems%20%281%29.en.html>

Glossary

SNR Signal to Noise Ratio